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Pratik Shrivastava

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SUMMARY

Aspiring Software Engineer with a solid foundation in programming, data structures, algorithms, databases, and backend development. Experienced in building real-world applications using Python, JavaScript, FastAPI, SQL, and Git. Eager to leverage technical and analytical skills to develop scalable software solutions and deliver business impact.

WORK EXPERIENCE

Software Engineering Intern — Siemens Technology & Services Pvt. Ltd.

May 2024 — Aug 2024

- Developed a near real-time frontend interface for an IIoT-based water monitoring platform, visualizing live consumption data from distributed sensors through interactive dashboards.
- Integrated REST APIs and implemented near real-time data updates, enabling operational monitoring and historical consumption trend analysis.
- Built GIS-based map visualizations with live meter status indicators and alerts, improving visibility into device health and flow anomalies across multiple locations.

Data Science Intern — Dodge AI

June 2025 — July 2025

- Designed and developed an end-to-end machine learning pipeline for customer purchase behavior prediction using regression models and Random Forests.
- Built a hybrid recommendation engine integrating content-based and collaborative filtering approaches to generate personalized product recommendations.
- Conducted exploratory data analysis and engineered behavioral and transactional features to enhance prediction accuracy and uncover customer purchasing patterns.

PERSONAL PROJECTS

Drape

[Project Link](#)

- Built an AI-powered visual fashion search engine that finds visually similar catalog products from an uploaded photo, using **YOLOv8** for garment detection, **DINOv2** for vector embeddings, and **FAISS** for similarity search.
- Designed a **neighbor-consensus** that derives garment descriptions and retailer search links directly from retrieval results – **avoiding a second model** on a constrained memory budget by making the description a direct readout of retrieval quality.
- Deployed the **FastAPI** backend to **AWS Lambda** as a container image, serving ~ 1-2s warm inference over a 20K-image catalog with free-tier limits.

HarmoniQ

[Project Link](#)

- Developed a hybrid recommender blending 4 retrieval sources – feature-space, similarity, audio, features, genre, and popularity – into one ranked score, with a per-result panel explaining each signal's contribution.
- Deployed the model (**FastAPI** on Render, **React/Vite** on Vercel, **Spotify OAuth**); also cutting cold start latency by preloading model artifacts on starting using a CRON job.
- Shipped surfaces on one artifact set – recommendations, mood discovery, genre exploration, playlist building, 3D PCA / feature-correlation visualization, catalog overview and a methodology page.

Booktunes

[Project Link](#)

- Built an AI-powered book illustration pipeline that extracts texts from PDFs via **OCR**, identifies characters using configurable **NER** (spaCy or single LLM pass per book) with union find alias merging, and generates page-consistent illustrations by feeding a running visual profile in each prompt.
- Designed a layer supporting multiple **LLMs** and image generation API, including **Groq**, **Gemini**, **Cloudflare Workers AI**.
- Deployed and maintained application on **Render**, **Vercel** and **Supabase PostgreSQL**.

SKILLS

Languages	Python, JavaScript, C++, SQL
Software Development	OOPs, Data Structures, RESTful APIs, JWT Auth
Databases	PostgreSQL, MySQL
Frameworks	FastAPI, Node.js, Express.js
Tools & Platforms	Git, Github, Docker (Basics)
ML & Data	Scikit-learn, Transformers, FAISS, YOLOv8, Pandas, NumPy, OpenCV

EDUCATION

Bachelor of Technology in Computer Science (B. Tech) — Manipal University Jaipur

July 2022 - June 2026